

DISTILLING RELIABLE PII DETECTION INTO A DEPLOYABLE REDACTION MODEL

Final Project Report



Group Based 5703 Capstone Project

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CONTRIBUTION STATEMENT

Our group CS45-1, taking project titled Distilling Reliable PII Detection into a Deployable Redaction Model , with group members JIAQI DANG, SIJIA LI, YUXIANG ZHOU, YINGNAN LIU, QICHEN ZHANG, SHIBO ZHANG, JUNDI HUANG, JIAXIN SHI, would like to state the contributions that each group member has made for this project during this semester:

- JIAQI DANG: Mainly responsible for the overall project promotion and coordination work, including organizing the project progress, standardizing the document structure, coordinating the division of labor among team members, and designing and maintaining the test set. I was also responsible for summarizing and analyzing the test results and participated in some implementation and verification work in the aspects of model lightweighting and fine-tuning experiments.
 - SIJIA LI: Primarily responsible for project meeting documentation, promptly responding to and coordinating communication with clients, as well as handling internal team coordination and equipment management. In the early stage, contributed to model selection and fine-tuning preparation. In the later stage, mainly worked on demo interface enhancement, testing, and overall presentation optimization.
 - JUNDI HUANG: Mainly worked on evaluation and testing design this semester. Contributions included reading literature on knowledge distillation and PII detection, developing testing standards and matching rules for generated data and model outputs, contributing to the milestones and references sections of the proposal, supporting server/model environment debugging, and later helping with teacher–student model comparison and evaluation metrics as the project moved into training and pipeline testing.
 - QICHEN ZHANG: Responsible for testing the design and preparing for verification, as well as writing the Resources section. At the early stages of the project, I contributed to discussions regarding PII detection results assessment. In addition to accuracy, I focused on span-level detection, sensitive labels, confidence values, and verification logic. The final OPF + Qwen + wrapper redaction system combines hardware, software, datasets, test resources, and team responsibility divisions.
 - YINGNAN LIU: .Mainly contributed to test data preparation, literature review, and data-format organization. Reviewed work on PII detection, de-identification, hard negatives, and uncertainty-aware redaction. Helped prepare test/evaluation data based on the Australian and University-specific PII taxonomy, covering realistic PII examples, ambiguous cases, and hard negatives. Also organized key data-format and validation logic to support OPF detection, Qwen classification, and wrapper-level decisions.
 - SHIBO ZHANG: Primarily responsible for training dataset quality assurance and the comprehensive analysis of system performance. In the early to mid-stages, I led the dataset acceptance testing, rigorously verifying the generated training data against
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the group's strict taxonomy standards to ensure high-quality inputs for model fine-tuning. As the project moved into the evaluation phase, I conducted in-depth error analysis and ablation studies (e.g., comparing OPF-only versus the Qwen hybrid pipeline). Ultimately, I authored the core "Results" section of the final report, synthesizing complex evaluation metrics (such as exact-match vs. overlap F1 scores) into clear insights regarding boundary misalignment, latency bottlenecks, and the system's overall effectiveness in handling ambiguous text.

- YUXIANG ZHOU: Focused on the final hybrid Australian PII extraction solution, combining OPF, Qwen, and policy-based rules. The work included preparing and cleaning JSON-span SFT datasets, validating entity offsets, and defining supported PII label types. It also involved fine-tuning and evaluating Qwen-based models, comparing alternative model approaches, and integrating model outputs with OPF and rule-based policy checks to improve reliability. Evaluation scripts were developed for cleaned and gold test sets, covering precision, recall, F1, difficulty-level analysis, and per-entity breakdowns. Shared redaction, parsing, and label-validation utilities were also implemented to support reproducible training, evaluation, and deployment.
- JIAXIN SHI: Mainly focused on two technical areas: privacy-preserving knowledge distillation and differential privacy. assisted the team in connecting with Microsoft Presidio, taking charge of synthetic data verification and early preparation for model distillation. Checked all generated private texts and multimodal data item by item, to guarantee all materials complied with format rules and met the project's quality requirements. Assisted in data rule optimization, evaluation metric validation and project presentation preparation.

All group members agreed on the contributions listed on this statement by each group member.

Signatures: JIAQI DANG (all group member signatures, either handwritten or digital signatures)

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ABSTRACT

This project aims to build a personal sensitive information identification and de-identification system tailored for the Australian scenario. It is designed to detect PII information in text, images, and PDFs, and to perform automatic masking, manual review, or ignore processing based on the risk level. The system adopts a hybrid architecture, using OPF/OpenAI privacy-filter as the high-recall candidate span detector, combined with Qwen9B span classification head for PII type determination, and then generating the final decision through the wrapper policy integrating model confidence, context information, format rules, and risk thresholds. The project also provides Fast API interfaces and web demos, supporting text input and file upload, and evaluating the performance of different backends in terms of precision, recall, F1, type accuracy, and latency using the test set.

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Note: The final report follows a similar structure to the proposal report, and reusing some content (e.g., literature review) is acceptable, but please revise the reused content to align with your final outcomes. Add any new materials that your group discovered during the project into the final report. Ensure that the overall similarity score of any report **is no higher than 35%**.

1. INTRODUCTION

As our daily digital information systems continue to improve and develop, we will find that increasingly personal sensitive information is appearing on online platforms, business systems, document records or log data. For example, names, email addresses, phone numbers, various ID numbers, medical insurance numbers, license plate numbers, bank account information, and home addresses, etc.

If these pieces of information are not correctly identified and de-identified during data sharing, model training, log analysis, or document processing, it is very likely to lead to privacy leakage and related compliance risks. Therefore, we hope to build a system that can automatically identify and mask PII, helping users better protect personal privacy security when processing text or file data.

This project mainly focuses on the PII identification and de-identification tasks in the Australian context. Compared to ordinary PII detection, the Australian scenario has some more localized and specific types of personal information. For example, TFN, which is Tax File Number, Medicare number, Australian mobile phone numbers, license plate numbers, as well as bank BSB and account numbers, etc.

Some of this information have relatively fixed formats, so they can be identified with the help of rules; however, they may also appear in different contexts, and their expressions may not be the same. For example, the same string of numbers could be a student ID, a bank account number, an order number, or even just an ordinary reference number in different contexts. Therefore, if only relying on a single model or only on fixed rules, there may be missed detections or false detections.

To solve this problem, we have designed a hybrid PII de-identification system. This system does not assign all tasks to a single model for completion; instead, it breaks down the entire process into several stages.

Firstly, the system will utilize the OPF model from the OpenAI open-source privacy-filter approach as the high-recall candidate span detector for the first stage. Its

primary function is to identify as many segments in the text that may contain PII as possible, that is, to answer the question "where might there be sensitive information" first.

Next, the system will use the span classification head related to the Qwen model to further classify and make probability judgments on these candidate segments. It will determine whether this segment is more likely to be a name, a mobile phone number, a student ID, a bank account, or actually not a PII.

Finally, the system will use the wrapper policy, which is the policy layer, to comprehensively consider the model's confidence, context information, fixed format rules, and risk thresholds to determine how each span should be ultimately handled. The system's decision is not simply a "yes" or "no" output, but is divided into three categories: automatic masking, manual review, or ignoring. The advantage of this approach is that in privacy protection scenarios, we can try to avoid missing out on truly sensitive information while delegating uncertain situations to human review for further examination.

At the same time, we have also developed a web-based demo and corresponding API services. Users can directly input text or upload files. For ordinary text, the system will directly proceed to the PII detection and de-sensitization process; for images or scanned PDFs, the system will first use VLM for visual text transcription to extract the text from the images, and then send it to the same set of PII detection process.

Overall, our project aims to achieve not just a single model experiment, but a relatively complete PII redaction pipeline. It includes model detection, type classification, rule judgment, risk decision-making, API invocation, and front-end presentation. The ultimate goal is to enable users to handle data containing personal sensitive information in a safer and more convenient manner.

2. RELATED LITERATURE

This project involves research related to PII detection, text de-tokenization, named entity recognition, LLM-assisted information extraction, model compression, confidence calibration, hard negative example learning, and deployable desensitization systems. It is necessary to carry out a literature review because this

project does not only focus on the presence or absence of sensitive information in documents. The implemented system must also be able to identify precise text spans, classify them into specific types of PII in the Australian and university scenarios, handle ambiguous values of similar identifiers, and return stable desensitisation decisions through an API and Web interface.

The literature reviewed in this project was selected from peer-reviewed journal papers, conference papers, and arXiv technical reports. The main contents include PII detection, text de-identification, named entity recognition, large language models, knowledge distillation, LoRA, confidence calibration, hard negative sampling, synthetic data augmentation, and redaction systems. The literature mainly focuses on articles published after 2015, but also covers some earlier de-tokenization work.

2.1 Literature Review

Early PII detection and de-tokenization systems usually rely on rules, dictionaries, regular expressions, and manually designed pattern matching methods. These methods work well when the identifier has a stable format, such as the email address, phone number, account number, or structured ID number in this project. Neamatullah et al. (2008) show that a rule-based and pattern-based approach is able to automatically remove identifying information from medical text records. Meystre et al. (2010) found that automated detokenization can reduce manual effort, but it is sensitive to format inconsistencies and context changes. This is relevant to the current project because Australian PII includes both fixed-format identifiers, such as TFNs, Medicare numbers, USIs, UAC IDs and BSB numbers, and context-dependent information such as salary, health status, addresses, academic records and demographic attributes.

Named entity recognition provides an important modelling foundation for PII detection because both tasks require span-level recognition. Lample et al. (2016) showed that neural sequence-labelling models can learn entity boundaries without heavy manual feature engineering. Later, BERT and transformer-based models improved contextual understanding through bidirectional representations (Devlin et al., 2019). This is important because the same string may have different meanings in different contexts. A number may represent a student number, a bank account number, an invoice number, a phone number, or just a common reference number. Instead of

relying only on fixed rules, this project uses OPF as a high-recall sequence labeling detector for locating possible PII text spans.

The importance of contextual span detection is further supported by Transformer-based de-tokenization. Johnson et al. (2020) applied a pre-trained bidirectional Transformer to a clinical de-tokenization task and showed that context models can improve the ability to identify spans of sensitive text. This important research shows that desensitization is not just a document-level classification problem. Document-level tags can only indicate the presence or absence of PII in a document but cannot identify the specific character positions that need to be obscured. In this case, the system needs to return the span offsets, PII types, confidence values, policy decisions, and finally the text after desensitization. Therefore, compared to simple document-level accuracy, span precision, span recall, type accuracy and overlap-based evaluation are more suitable for this project.

Recent LLM-based research has shown that large language models are able to support de-tokenization tasks by understanding context and following structured extraction instructions. Singhal et al. (2024) found that GPT-4 is able to detect a large number of sensitive entities in student texts, but may still over-redact or misjudge information due to different contexts. Yermilov et al. (2023) also point out that privacy-preserving NLP systems need to strike a balance between privacy protection and usability. However, the cost of directly deploying LLM is high. Kwon et al. (2023) found that LLM service performance is significantly affected by GPU memory, batching, and Key-Value cache management. These studies underpin the hybrid design of the project: LLM-based components are suitable for semantic judgment, but the final desensitization system should return controllable outputs rather than rely on free-form generation.

Knowledge Distillation together with Parameter-Efficient Learning provided the theoretical foundation for the model exploration early in this project. Hinton et al. (2015) proposed a knowledge distillation method for transferring valid behaviors from a large teacher model to a smaller student model. Hu et al. (2021) proposed LoRA to reduce the fine-tuning cost by training only low-rank adaptation matrices instead of all model parameters. These methods are relevant to our project, as teacher-student and Qwen based training routes were explored earlier in the project. The implementation

focuses more on the actual deployment: The Qwen backbone model is frozen and only the lightweight span classification head is trained, and the classification is done based on cached span representations. This allows Qwen to provide controllable probability estimates for the candidate spans detected by OPF.

Confidence calibration, hard-negative learning and synthetic data are important components of PII desensitization. Guo et al. (2017) pointed out that modern neural networks often suffer from insufficient probability calibration, and temperature scaling can improve the reliability of probabilistic outputs. Qwen's span-head outputs probability distributions including NON_PII, and the policy layer uses these probabilities to decide whether a span should be desensitized, manually reviewed, or ignored. Xu et al. (2023) show that better sampling of negative samples can help the model distinguish real entities from similar non-entities, which provides support for the design of hard negative examples in this project. Kim et al. (2024) show that privacy-secure data augmentation methods can support de-tokenized model training while reducing the risk of disclosure of real sensitive information. These studies support the design of this project using synthetic examples related to Australian scenarios, university scenario templates, teacher-generated data, controlled formats, as well as hard negative examples.

Overall, reliable PII desensitization requires a combination of approaches. Rule-based methods are suitable for identifiers with stable format. Sequence labeling model can help locate sensitive text span. The LLM-based components can improve the ability of context semantic judgment. Confidence calibration helps to support risk-aware decision making. Hard negative examples reduce false positives. This project integrates these ideas into a deployable Australian PII desensitization pipeline. These include OPF candidate detection, QWEN-based probabilistic scoring, deterministic post-processing, policy-based decision mechanism, file processing, schema validation, FastAPI interface, and Web Display functionality.

3. PROJECT PROBLEMS

The central question of this project is how to build a reliable and deployable PII desensitization system for sensitive information in Australian and university scenarios. Although LLM can understand the context and identify sensitive information, if it is

directly used for all desensitization tasks, it will face problems such as high cost, difficult to control, and not suitable for stable large-scale deployment. Relying only on rule-based systems also has limitations, as many PII-like values can only be correctly interpreted in context.

The key of this project is not just "how to identify PII". The system must also determine where the sensitive text span is located, what type of PII it belongs to, how reliable the prediction is, and whether the value is PII or hard negative. And whether it should ultimately be desensitized, manually reviewed, or ignored. The final scheme adopts a hybrid architecture: OPF is responsible for detecting candidate spans, Qwen re-scores them using class probabilities, and the policy layer is responsible for making the final desensitization decision.

3.1 Project Aims & Objectives

The main objective of this project is to develop a deployable Australian PII desensitization system that is able to identify sensitive text spans, classify them into detailed classification systems, and support risk-based decision making. Our system aims to reduce reliance on direct inference from large models while leveraging LLM-based semantic information where applicable.

The first goal was to maintain a consistent taxonomy of PII for the Australian and university scenarios. The project employs an OPF label space with 79 categories, and an 80-class Qwen label space with NON_PII. The taxonomy covers common identifiers such as name, date of birth, email, mobile phone number, address, student number, UAC ID, USI, TFN, Medicare number, passport number, BSB number, bank account number, and bank card information. It also includes sensitive categories such as medical information, disability or health conditions, Aboriginal and Torres Strait Islander identity, ethnic background, criminal records, salary, academic performance, WAM, social media identifiers, geolocation and biometric data. The result is that these categories are consistently used across model outputs, policy rules, data schemas, and ultimately desensitizing responses.

The second goal is to construct an effective process for candidate span detection and classification. OPF is responsible for scanning the input text and locating possible

PII text spans, while Qwen's Span-head classifies each candidate span into an output space containing 80 categories and generates a calibrated probability distribution. This process enables the system to express uncertainty when a certain value may belong to more than one category, or is not PII. This objective is evaluated by span-level precision, recall, F1 score, offset correctness, classification accuracy, top-k accuracy, macro-F1, and calibration quality were evaluated.

The third goal is to support risk-based desensitizing decisions and reduce false positives. The policy layer combines model confidence, context information, deterministic rules, and risk thresholds to decide whether each span should be desensitised, manually reviewed, or ignored. This metric is important because privacy risk is not the same as model confidence. By using hard-negative examples and NON_PII categories, the project retains the real PII detection ability while reducing unnecessary masking.

The fourth objective is to deliver and evaluate a usable desensitization service. The system supports FastAPI interface, Web presentation, structured JSON output, schema validation, policy configuration, deterministic desensitizing text output and file input functions. For text input system, OPF + Qwen + policy processing pipeline will be directly applied. For images or scanned PDFs, however, text extraction or transcription is performed before the desensitization process is applied. The system is evaluated through model metrics, calibration checks, wrapper tests, smoke tests, hard-negative examples, API behaviour and output validity.

3.2 Project Questions

The main project question is: How can a large-model-assisted PII detection capability be converted into a deployable and uncertainty-aware redaction system for Australian and University-specific sensitive information?

This main question can be divided into several sub-questions.

First, can the system accurately detect PII spans across the defined Australian and University-specific taxonomy? The system must handle both fixed-format identifiers, such as emails, phone numbers, Medicare numbers, student IDs, TFNs and bank details, and context-dependent information, such as salary, gender, medical information, addresses and geolocation.

Second, can a hybrid architecture improve the balance between recall and precision OPF provides high-recall candidate detection and stable character offsets, while Qwen provides semantic probability estimates for candidate types. The project investigates whether this combination is more practical than rule-only detection, OPF-only detection, or free-form LLM generation.

Third, can the system represent uncertainty and reduce false positives? Many values have multiple possible meanings. A date may be a date of birth, expiry date, or ordinary deadline. A number may be a student ID, bank account number, invoice number, or general reference number. The project examines whether top-k probabilities, NON_PII classification, calibration, context windows and post-processing rules can support safer redaction decisions.

Fourth, can the model components be integrated as an actually usable service? The final deliverable should not only contain experimental results, but also provide a stable API, a working Web interface, predictable JSON output, file upload support, deterministic desensitization results, and a clear configuration mechanism. So service integration is also part of the problem of this project.

3.3 Project Scope

The scope of this project is limited to PII desensitization in the Australian and university scenarios. The system focuses on the taxonomy defined in the project, which includes government identifiers, payment information, contact information, academic records, employment and income related data, medical information, sensitive demographic information, online identifiers, geolocation, and biometric data. Our project does not cover all international identifier formats, and defines privacy categories outside the taxonomy.

The project only uses privacy-safe data. Real University personal information is not used for model training. Data pipelines rely on synthetic examples, teacher-generated samples, controlled templates, hard negative examples (negatives) and validation scripts. This approach reduces privacy risks during development and enables repeatable experiments.

The core input type of the system is text. The system also supports text files, PDF files and image files. File support is an input extension to the text redaction

pipeline. For PDFS containing an embedded text layer, the system reads the text directly. However, for images or scanned PDFS, text extraction or visual transcription is required first, and then the extracted text is fed into the same OPF + Qwen + policy pipeline for processing. Complete document layout understanding, handwriting recognition, and complex multimodal reasoning are outside the scope of this project.

The project delivered a working prototype, not a full production-grade system. The prototype includes FastAPI interface, Web interface, configuration management, policy setting, schema validation, testing capabilities, file upload support, and deterministic desensitizing output. Production-grade identity authentication, cloud infrastructure integration, long-term monitoring, and formal legal compliance are not part of the scope of this project.

The project supports risk-aware redaction, but there is no guarantee that all personal information will be completely removed. The system is able to automatically desensitize high-confidence content and submit uncertain cases to human review. Human inspection is still important for ambiguous or high-risk scenarios. It is essential for more difficult categories such as gender, latitude and longitude information, ambiguous dates, Medicare numbers, and information related to BSB/ bank accounts.

Overall, the project was feasible because it followed a clear and testable process. OPF is responsible for detecting candidate spans, Qwen re-scores candidate content using calibrated probabilities, policy rules decide whether to desensitise or manually review, and wrapper returns structured JSON and processed text. The scope of the project addresses current customer issues while leaving wider international coverage, production integration, and advanced multimodal document understanding as future work.

4. METHODOLOGIES

4.1 Methods

This project used data science methodology to develop an automatic Australian Personally Identifiable Information (PII) detection and redaction system. The final system is a three-stage hybrid architecture: an OPF token-classification model for high-recall candidate span detection, a Qwen3.5-9B span classification head for PII

type probability estimation, and a wrapper policy layer for final REDACT, REVIEW, or IGNORE decisions.

The system was implemented in the following phases: dataset preparation, model training, model evaluation, wrapper integration, frontend/API implementation, and final validation. The proposed approach was selected as PII redaction is a risk-sensitive task where new errors found at evaluation should be fed back to data cleaning, threshold tuning, and policy design. Rather than one generative model across all modules, the project decouples span detection, type classification, and policy decision. This improves explainability, simplifies identification of errors, and diminishes the possibility of unstable free-form model output.

The hybrid approach was chosen because OPF is effective and appropriate for discovering possible PII spans, while Qwen offers superior semantic judgement for classifying candidate spans. The wrapper policy is subsequently informed with deterministic rules, confidence thresholds, context checks, and human-review logic. The latter design is well suited for privacy protection by allowing uncertain cases to be escalated to a review, instead of being ignored.

4.2 Data Collection

This dataset was derived from an augmented Australian PII corpus with annotated text records from multiple document styles, including forms, emails, clinical-style notes, letters, records, system logs, payslips, and narrative examples. The dataset also featured hard-negative records and candidate-level negative samples to enable the model to discriminate true PII from the normal text that might resemble PII.

The final label space had 79 Australian PII classes for OPF span detection, including PERSON, DATE_OF_BIRTH, EMAIL_ADDRESS, MOBILE, ADDRESS, AU_TFN, MEDICARE_NUMBER, STUDENT_ID, UAC_ID, and USI. For the Qwen span classifier, one additional NON_PII class was added to that and led to 80 candidate-level classes.

The data were split into train, development, and test sets using a group-key hash split process. This was used to lower the risk of data leakage as it prevented the duplication of templates or records in the same format between splits. Additionally,

offset validation was used to verify that each annotated span matched the substring in the source text.

For OPF training, the dataset contained:

Splits	Records	PII Spans	Empty-span Hard-negatives
Train	81,298	109,977	64,387
Development	10,034	13,412	7,977
Test	9,659	13,513	7,557

For Qwen span-head training, the cached candidate-span examples were:

Splits	Span Examples
Train	113,354
Development	13,850
Test	13,588

This sample size was considered sufficient for statistically meaningful evaluation because the test set contained thousands of records and more than 13,000 labelled spans across the full PII taxonomy.

4.3 Data Analysis

This data analysis process involved model performance, class-level errors, confidence calibration, and final redaction behaviour.

Firstly, OPF was first performed training as a token-classification and sequence-labelling model. It aimed for high recall detection of potential PII spans. OPF output was determined based on token accuracy, detection precision, detection recall, detection F1, span precision, span recall, and span F1. OPF demonstrated efficient behaviour as the first-stage detection across these samples, with a detection F1 of 0.9793 on the test set, followed by a span F1 of 0.9725 on the test set.

Second, the Qwen3.5-9B backbone served as a frozen feature extractor. Span embeddings were cached from the model, and some lightweight classification heads were compared, such as mean pooling, first-token, last-token, and concatenated MLP heads. We selected the last_linear classifier as the final head because it possessed the best development negative log-likelihood.

The Qwen span-head trained applied soft cross-entropy, with weighted hard negatives, and higher weights for NON_PII, since false positives are common when normal text look similar to identifiers. The Qwen span-head received a Top-1, Top-3,

negative log-likelihood, micro F1, macro F1, and weighted F1. On the test set, the latter Qwen 9B hard negative span-head obtained a Top-1 accuracy and a Top-3 accuracy of 0.9875 and 0.9998, respectively. The macro F1 was 0.9755, which meant the test achieved steady performance for common and uncommon classes.

Lastly, cover the wrapper policy that took into account model confidence, Top-k ambiguity, span format on a continuum, contexts surrounding, deterministic pattern evidence and PII sensitivity level. And that was the very last system decision that the layer delivered: ‘Redact high-confidence spans automatically, assign uncertain ones to be reviewed by humans and not include clearly non-PII spans’. We opted for this method in the past, as privacy redaction has asymmetric risks (it is more serious to miss true PII than send unnecessary cases under review).

4.4 Deployment

The final system is deployed through a FastAPI-based redaction wrapper. The wrapper exposes API endpoints for health checking, text redaction, file redaction, and a browser-based demo interface. The deployment configuration uses the hybrid backend configuration file and policy configuration file, allowing the backend model and policy thresholds to be updated without changing the API layer.

The deployed pipeline accepts direct text input, text files, PDFs, and images. Text files are decoded directly, PDFs with embedded text are parsed through their text layer, and scanned PDFs or images are transcribed before being passed into the same OPF-Qwen-policy pipeline. The final output includes detected spans, predicted PII types, confidence information, redacted text, and review decisions.

Updates and bug fixes are distributed through versioned code, backend configuration files, policy files, and model artifacts. Qwen span classification heads are stored as lightweight model artifacts, while larger OPF and Qwen trunk checkpoints are prepared separately on the deployment server. This separation makes the system maintainable because policy thresholds, deterministic rules, and frontend/API behaviour can be updated independently of large model weights.

4.5 Testing

To ensure the quality of the model and the reliability of the software multiple levels of testing were performed. In the model-level analysis, OPF span detection and

Qwen span classification were tested separately. OPF testing was conducted on development and test sets using detection F1 and span F1. Qwen was validated with Top-1 accuracy, Top-3 accuracy, negative log-likelihood and F1 metrics. In order to check weaker classes such as GENDER, LATITUDE, LONGITUDE and SALARY_WAGE_EXPECTATION, additional per-label analysis was also performed.

Integration testing was performed in the overall hybrid pipeline to make sure OPF spans could be passed successfully into the Qwen span-head and handled by the wrapper policy. Representative smoke tests covered dates of birth, email addresses, phone numbers, addresses, salary values, passport numbers, ID numbers, personal names and other negative examples. These tests ensured that the correct REDACT, REVIEW, and IGNORE decisions had been made in the entire workflow.

More tests included frontend, API schema, OCR/file-input components, redaction utilities and training pipeline as part of automated Python tests. Manual web-interface testing was also conducted on various text inputs to compare actual user interaction behaviour. Accuracy levels were obtained for each of the categories input for data including passport numbers, ID numbers and names so that the reliable frontend detection performance can be checked.

In conclusion, the combined test strategy tested not only the predictive strength, but also the reliability of the system and confirmed that the implemented PII redaction system is stable and reliable under actual use conditions.

5. RESOURCES

This section describes the main resources used by our team when completing the project. Since this project involves not only training a model but also data construction, model testing, backend implementation, web demo, file processing, testing, documentation organization, and team collaboration, it requires a considerable number of resources. In the final system, OPF is used to identify the spans that may contain PII, Qwen is used to classify these spans, and wrapper provides the final REDACT,

REVIEW or IGNORE decisions. Therefore, these resources need to support both the machine learning part and the deployable software part.

5.1 Hardware & Software

The daily development work is mainly carried out on the team members' laptops. Windows, macOS and Linux devices are all used for code writing, dataset checking, document editing, GitHub updates and small-scale testing. Since the wrappers and APIs are mainly written in Python, if the corresponding Python environment and dependencies are installed, this part of the system can run on common operating systems.

Model experiments require more computing resources. The components related to Qwen are relatively large, and regular laptops are difficult to run smoothly, especially when generating span embeddings or conducting large-scale evaluations. Therefore, model testing and span-head experiments require devices with GPU support. In the final design, the large model backend and wrapper, as well as the demo, are separate. The testing process is simplified because modifying the model or threshold does not require re-editing the entire front-end or API of the application to change it.

The software part of the project is organized around the tasks associated with the project. Python is the main development language. PyTorch and Hugging Face Transformers are used for modelling experiments and Qwen span-head. Data inspection, label processing, and evaluation results analysis are all carried out with the help of Pandas, NumPy, and scikit-learn. OPF/OpenAI privacy-filter serves as the first-stage detector, designed to locate potential PII spans before QWen further classified.

The FastAPI connects user input with the editorial pipeline and returns structured output. The web demo is built using HTML, CSS and JavaScript. Users can enter text, upload files, and view detected spans, confidence, decision, and redacted text. The project will also require OCR or VLM-related tools to be able to convert the images and scanned PDF documents into text before they are entered into the same pipeline as the text files.

GitHub manages code, dataset scripts, model-related files, wrapper code, API files, configuration files, and report materials throughout the entire project process.

This is crucial because there are multiple interrelated parts to the project. For instance, changes to labels, policy thresholds or backend settings can all affect model testing and demo performance.

Overall, the hardware and software resources are arranged according to the final prototype system structure. The OPF requires labeled span data and detection scripts; the QWEN requires GPU support and cached span representations; the wrapper requires policy files and thresholds; the demo requires APIs and front-end resources. These resources help the team advance the project from the model idea to a functioning redaction system.

5.2 Materials

During the semester, Australian and university-specific PII datasets were constructed. It includes synthetic text records, labelled PII spans, hard-negative examples, and candidate-level records for Qwen span heads. Since this was a privacy protection project, the team did not use real university personal information for the training. There would be an increase in privacy risks if actual personal information was used.

In this dataset, we have included a wide variety of text styles, such as tables, emails, letters, system logs, record files, similar clinical records, similar payroll samples, and narrative texts. This is necessary because PII can appear in many scenarios. It is possible to find information such as the Tax File Number, Medicare number, student ID number, phone number, address, and bank details on table fields, in emails, or in ordinary sentences, such as in emails.

As part of the dataset, hard-negative examples have also been added. Values that resemble PII (Personally Identifiable Information) but should not be anonymized, such as invoice numbers, order IDs, reference numbers, common dates, course codes, public URLs, and other non-sensitive identifiers. These are very useful because the system cannot simply cover up all the content without revealing the numbers or similarly formatted dates for the user to see. Tests can be conducted using these samples to see if false positives are reduced.

The project also used label mapping and taxonomies. There are 79 categories in the Australian PII label space that are used by OPF. QWEN span classifier used an 80-

category label space with an additional NON_PII category. The mappings ensure consistency between datasets, model outputs, policy rules, API responses, and final redactions.

The project used configuration and testing materials, including backend configuration files, policy settings, confidence thresholds, rules, schemas, model paths, train/development/test splits, smoke tests, API tests, wrapper tests, file upload examples, and representative text inputs. The materials were used to check whether the system could return the correct span, PII type, confidence value, and redacted text.

Additionally, the team has also retained weekly notes, meeting minutes, client feedback, updates from GitHub, summaries of experiments, screenshots of demos, evaluation tables, and presentation materials. It is important for our team to keep track of the progress of the project using these records. There is a reason why the final direction of this project has shifted from the earlier idea of distillation to the OPF, QWEN, and Wrapper pipelines in place of distillation.

In addition to the risk associated with synthetic data, there is also the possibility that it might be cleaner than real data. For example, if the samples are too simple, the system might perform well during testing, but it tends to deteriorate when it is faced with real-world, ambiguous scenarios. This issue was mitigated by incorporating severe negatives, expanding the taxonomy, examining span offsets, using separate development and test sets, and combining the model output with wrapper decisions.

5.3 Roles & Responsibilities

Various parts of the project were divided among the team. It included project coordination, literature review, dataset construction, model work, evaluation, wrapper policy, API and demo implementation, testing, documentation organization, and customer communication as well. During the semester, the project direction changed. At first, the team considered a knowledge distillation approach, but later it turned out that the work was focused more on the development of a deployable OPF + QWen + wrapper pipeline, so some tasks had to be adjusted as well.

Jiaqi Dang is responsible for coordinating the overall project schedule, including task arrangements, weekly progress tracking, reporting structure organization, and support for test sets and result summaries as well as ensuring that they are competitive.

Also, Jiaqi was involved in the discussion about how to make models lighter and how to fine-tune them.

Sijia Li was involved in the preparation of meeting minutes, customer communication, team coordination within the team, and equipment related tasks as well as providing demo support. As part of the later stages of the project, Sijia assisted in improving the demo interface, preparing test materials, and preparing presentation materials for the demo.

Yuxiang Zhou is one of the people who contributed to the creation of the Personal Identifiable Information dataset, having prepared examples of PII that are relevant to Australia, as well as supplementing the data with different formats of PII. As a result of this work, the coverage of training and testing data has been enhanced.

Yingnan Liu participated in the literature review, project problem analysis, and lightweight model's research. Yingnan assisted the team with knowledge distillation, model limitations, confidence-related designs, and review logic.

Qichen Zhang played a critical role in the evaluation of design support. He also prepared details for verification and drafted the Resources section in the final report. Qichen participated in discussions about PII recognition. The focus was on span-level detection, sensitive labels, confidence values, and verification logic, rather than simply accuracy. Qichen also organized the idea of comparing generated data and model outputs. Qichen summarized the hardware, software, datasets, tests, and team role resources required for OPF + Qwen + wrapper at a later stage.

Zhang Shibo worked on data augmentation and quality inspections of synthetic data, as well as LLM-as-a-Judge. During this part of the work, the team was able to think about how to check the generated data according to testing standards, and how to use the teacher model outputs to support other tests later.

Jundi Huang mainly participated in the organization of the GitHub repository structure, the preparation of the training environment, and the related evaluation work. Specifically, he helped organize the code files, configured the model running environment, supported the span checking logic and threshold analysis, and participated in the later model comparison.

Jiaxin Shi participated in the Presidio-related research, the synthesis of data sets for verification, and the multimodal/file-input checks. Jiaxin investigated how rule-based recognizers and existing PII detection tools could support the system and also helped verify image/file samples and testing thresholds.

Overall, the team roles cover the main resources required for the final prototype. Data-related work supports the training and testing materials; model-related work supports OPF, Qwen and evaluation; implementation-related work supports the wrapper, API and demo; coordination-related work supports communication, documentation and milestone tracking. As a result of this division of Labor, the team can manage project risks and deliver a functioning prototype system to the client, rather than just a conceptual model.

6. MILESTONES / SCHEDULE

Milestone	Tasks	Reporting	Date
Week-1	Initial project familiarisation and discussion of project direction.	Internal group discussion.	8-03-2026
Week-2	Review the relevant literature on knowledge distillation and privacy-preserving natural language processing; clarify the overall technical direction. In the initial stage, the focus was mainly on exploration and concentrating on the teacher-student style knowledge distillation to achieve the de-identification of personal identity information.	Client meeting to review the work plan	15-03-2026
Week-3	Narrow down scope, determine high-level PII types, start drafting synthetic dataset creation process and GitHub/environment setup. During this phase the team determined that we would move forward with building a lightweight PII detection/masking system. We also settled on building an 8000 item synthetic data set.	None	22-03-2026
Week-4	Establish testing standards, generate synthetic data, and prepare for later comparison	None	29-03-2026

	between generated data and teacher model outputs. Compared with the original proposal timeline, the project moved more quickly into dataset and evaluation design because client feedback clarified the workflow.		
Week-5	Proposal Report Due		
Week-6	Dataset setup/verification/validation, along with tech setup continued past proposal submission. Ended up being a sort of limbo phase between the planning of proposal and experimentation. Some anticipated forward movement on validating models against each other was hindered by environment setup and technical glitches.	Client meeting to review GUI	12-04-2026
Week-7	Develop distillation pipeline end-to-end: preprocessing + teacher supervision + converting labels + training student + early validation. This is a significant shift from planned proposal timeline, as by this point the group was already doing pipeline-level experiments vs. only preparing dataset.	None	19-04-2026
Week-8	Did more model-related experimentation and noticed that simple evaluation metrics were still not granular enough to diagnose performance correctly. Served as a bridge week between validating my pipeline and doing in depth model analysis.	None	26-04-2026
Week-9	Progress Report Due		
Week-10	Reconstruct the training and testing datasets so that one	Client meeting to deploy the system	10-05-2026

	input can be linked to multiple possible PII dimensions and confidence scores. The group also tested the OPF + Qwen hybrid pipeline and achieved strong initial results. This marks a shift from simple label prediction to probability-distribution-based classification.		
Week-11	Retrain with this new dataset or use for better aligned evaluation. The team swapped to Qwen3-VL-8B-Instruct model since they thought OCR was limiting performance too much. But accuracy fell to approximately 85%, so more fine-tuning will have to occur.		17-05-2026
Week-12	Final Presentation		
Week-13	Final Report (thesis)		

7. RESULTS

7.1 Results and Deliverables

One of the key objectives of this project was to develop a highly accurate, deployable PII redaction system that didn't depend on huge, high latency models. To answer this, the project ended up with a number of important outputs.

First, we came up with a 79-class Australian context-specific PII taxonomy which included Government IDs, payment information, and medical records. We then managed to effectively train a span detection model (the OpenAI version Privacy Filter) and a relatively light classification head (Linear, 1.3 MB, representation Qwen3.5-9B-Base). These models were then fitted in a full redaction pipeline. Detected text is handled by a deterministic pipeline, which uses redact, review, or ignore rules. Lastly, the complete system is available via a web service implemented with FastAPI, also with a user interface and automatic tests.

The performance of every model component is described and how each helps solve the major challenges in privacy and latency for the project.

7.2 OPF Span Detection Results

Overall Detection Efficacy

To determine whether the OPF model would cope with the tricky Australian PII types, we examined its performance overall (Table 7-1). The test-set F1 score was 0.979. It was able to identify approximately 13300 out of the 13513 actual PII spans successfully. In other words, it doesn't blindly guess and generate false alarms on the normal text – that's the high precision score (0.975). The span F1 score examines exact boundary alignment. On here the score was 0.973, which is a very high score, meaning that the model is excelling at detecting the precise start and end of words prone to sensitivity.

Table 7-1. OPF span detection metrics

Split	Token Acc.	Det. Prec.	Det. Recall	Det. F1	Span Prec.	Span Recall	Span F1
Dev	0.989	0.963	0.982	0.972	0.944	0.978	0.961
Test	0.992	0.975	0.984	0.979	0.968	0.977	0.973

Class-Level Performance Analysis

On closer inspection of each type of PII, the responses were quite diverse (Table 7-2). The model was very successful with the 47 classes classified as "strong" e.g., major government IDs, bank IDs etc., and scored virtually all the points in these classes.

While the weaknesses in the "weak" group indicate a great challenge, however;—text ambiguity. Classes without a fixed format struggled significantly. Examples of this are the GENDER class, which missed numerous targets due to the inclusion of the word "male" in all other words in normal text. The SALARY_WAGE_EXPECTATION on the other hand, almost got everything, but falsely classified too many (precision 0.535). It is clear from this trade-off that the OPF detector is quick and this is where we need another classification stage to pull out the hard to classify but rich in context words.

Table 7-2. Per-class span F1 on the test set, grouped by performance tier

Tier	Count	Representative classes (with F1)	F1 Range
Strong ($F1 \geq 0.95$)	47	AU_TFN 1.00, MEDICARE_NUMBER 1.00, PASSPORT_NUMBER 1.00, STUDENT_ID 0.99, USI 0.98, MOBILE 1.00, IP_ADDRESS 0.99, PERSON 0.99, DATE_OF_BIRTH 0.96	0.95–1.00
Moderate ($0.70 \leq F1 < 0.95$)	7	GEOLOCATION_INFORMATION 0.87, CREDIT_CARD_EXPIRY 0.86, NUMBER_PLATE 0.91, PAYMENT_CARD_NUMBER 0.89, CONTRACT_TYPE 0.93, EMAIL_ADDRESS 0.90, NATIONALITY 0.92	0.86–0.93
Weak ($F1 < 0.70$)	4	SALARY_WAGE_EXPECTATION 0.69 (prec. 0.535, rec. 0.986), LATITUDE 0.52 (prec. 0.607, rec. 0.447), GENDER 0.36 (prec. 0.591, rec. 0.260), LONGITUDE 0.31 (prec. 0.857, rec. 0.188)	0.31–0.69

7.3 Qwen Span Classification Head Results

Model Selection and Calibration

We took the features from the frozen Qwen model, and we then tested four different classification heads to find the one that provides the optimum combination of accuracy and size. The last_linear version, as can be seen in Table 7-3, was the most successful on our development set and so we selected last_linear for our final system. It requires only 328,000 parameters. Illustratively, the significantly bigger concat_mlp version (5.4 million parameters) performed worse. Also, the bottom line was last_linear with a scaling factor of 1.005. Because this is just 1.0, the probability value(s) of the model were already pretty good and have not needed any finer tuning.

Table 7-3. Head architecture comparison on the development set

Architecture	Dev NLL	Temperature (post-scaling)	Dev NLL (post-scaling)
mean_linear	0.124	1.165	0.147
first_linear	0.149	1.324	0.183
last_linear	0.085	1.005	0.085

concat_mlp	0.105	1.137	0.124
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Performance Consistency and Error Analysis

The results obtained in terms of top-1 accuracy remained virtually stable throughout training, development and test sets (see Table 7-4). This indicates that the train technique was able to prevent overfitting.

We observed some useful lessons in the way we were going wrong and what the limits of the system were. In the general NON_PII class, the model performed well, always making the same top-1 prediction but with a fairly high ECE (0.333). The model might be too confident or even fail to be confident when its guess is correct for NON_PII. When we tried testing the system with those away from the “ambiguous” examples we made before, it was 0.779. Rather than many failures, it serves as a reminder for us that these type of cases that are dependent on context is where the model is at its current boundaries.

Table 7-4. Final classification head metrics (last_linear, temperature 1.005)

Split	Spans	Top-1	Top-3	Micro F1	Macro F1	Weighted F1	NLL
Train	113,354	0.986	1.000	0.986	0.974	0.986	0.073
Dev	13,850	0.986	1.000	0.986	0.974	0.986	0.085
Test	13,588	0.988	1.000	0.988	0.976	0.987	0.077

7.4 Hybrid Pipeline: Integration and Ablation

Pipeline Integration and Ablation Study

We wanted to see how helpful the Qwen model was, so I switched to older versions of the pipeline for testing (Table 7-5). The 0.881 is the exact recall for the 9B model.

What caused this to occur? While the OPF detector is able to determine where sensitive data is in a reasonable location, it has poor detection of exact character boundaries (overlap F1 of 0.882). The introduction of the Qwen classification stage corrects such misboundary errors. When there is an overlap between the detector and the real data, the detector will be instructed by the Qwen head on 96.7% of the occasions. That's proof that our two stage design makes a difference.

Table 7-5. End-to-end pipeline comparison across backend configurations

Backend	Exact P	Exact R	Exact F1	Overlap P	Overlap R	Overlap F1	Type acc.
OPF only	0.725	0.772	0.748	0.858	0.908	0.882	—
9B hybrid	0.732	0.881	0.799	0.815	0.976	0.888	0.967
4B + OPF bias	0.740	0.805	0.771	0.871	0.943	0.906	0.958

Smoke Testing Analysis

Finally, we conducted a smoke test using ten tricky-inputs that were handcrafted (Table 7-6) to observe the rules in the real world. It successfully caught standard identifiers like emails. What it did really well was with poorly-formed sentences, such as “salary \$85,000”; the risk score was 0.460. This is because this system didn't merely stow it away, it recognized that it would need to be reviewed by a human, so it is sent there. But it totally missed a very poorly typed BSB and Medicare number! This confirms a known problem that if the format is completely broken then our detection rules may not work.

Table 7-6. Smoke test results across ten hand-crafted inputs

Input	Expected type	Detected	Predicted type	Disposition	Risk
DOB: 04/05/1998	DATE_OF_BIRTH	Yes	DATE_OF_BIRTH	review	0.484
Deadline 15/06/2025	DATE_OF_BIRTH	No	—	—	—
BSB 062-000 acct 12345678	BANK_ACCOUNT	No	—	—	—
alex@example.com	EMAIL_ADDRESS	Yes	EMAIL_ADDRESS	review	0.420
412 345 678	MOBILE	Yes	MOBILE	review	0.264
2 Wallaby Way, Sydney	ADDRESS	Yes	ADDRESS	review	0.440
Order ORD-987654	(negative)	Yes	—	—	—
identifies as male, he/him	GENDER / PRONOUN	Partial*	PRONOUN	ignore	0.000
salary \$85,000	SALARY	Yes	SALARY	review	0.460
Medicare 2123 45678	MEDICARE	No	—	—	—

*The OPF detector identified "he/him" (correctly classified as PRONOUN) but missed "male" (GENDER).

7.5 Deployment Metrics

Deployment Performance and Latency

We then measured metrics of the deployed FastAPI service to find out whether the lightweights met our latency requirements in the real world or not. It takes about 142 seconds to initialise the server and consumes 10 GB of the GPU. Running, however, the system can handle standard text requests of 60 characters in only 43 milliseconds.

Uploaded files take longer to process, depending on the file's format. Embedded text PDFs are scanned in milliseconds whereas document pages require visual-language translation which takes several seconds to process per page. In the overall picture the above latency results showed that the system is very well suited for real-time text-pre-processing in general.

7.6 Implications and Challenges

Implications and Ongoing Challenges

The final pipeline directly solves the primary problem within this project which is processing Australian PII safely, without requiring huge pieces of computing power. The most efficient feature is the 3 way policy (redact, review, ignore). That one prevents the system from simply letting some more ambiguous text flow by, a very important piece for university and medical environments.

But when we examined the results a few problems cropped up that we need to work on in the future:

1. Patterns have more to play than the size of the data: Many patterns showed that increasing the amount of data didn't correlate with increasing accuracy. All we did was give the class CENTRELINK_REFERENCE_NUMBER 727 training examples and gave it a perfect F1 score. GENDER's performance was, in contrast, not that good. The dataset is much less useful than having a pattern when finding IDs, as demonstrated above.
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2. The OPF was the biggest practical problem - speed limits. Running it on its own increased the latency per request by +0.9 second. As a result, we had limited time to perform a full end-to-end evaluation on all 9659 examples on the test set prior to the end of the project.
3. Rigorous security settings: Our security thresholds are very conservative, to be on the safe side. In our testing we did not see any automated redaction; everything was taken to a human review. This is safe to do, but a physical product would have to tweak these numbers to find a balance between automation and people doing the checks.

8. DISCUSSION

This section summarises the key findings, discusses the practical and academic implications, identifies contributions to existing research gaps, provides actionable implementation recommendations, and reflects on the project iteration from Week 2 to Week 12.

8.1 Core Results and Project Evolution

We built a combined PII masking system. It meets the privacy rules for local areas and universities in Australia. Realworld tests show it runs without any issues. Our workflow has two stages. First, we used the OPF span detector. It gives a detection F1score of 0.9793 and a span F1score of 0.973. Secondly, we used a lightweight classification part called Qwen3.59B.

The main network of this model remained frozen throughout the whole process. It supports 80 types of classification in total. The Top-1 accuracy reaches 0.9875, and the macro F1-score stands at 0.9755. We carried out important and practical revisions to our technical plan from Week 2 to Week 12. In the early stage of the project, from Week 2 to Week 3, we mainly looked into the traditional teacher-student knowledge distillation framework. Still, a single end-to-end generative model works poorly in real privacy protection tasks. It costs much to deploy. Its output is unstable, and detection boundaries are also hard to tell apart.

To solve these problems, we made a clear switch to our technical solution between Week 7 and Week 10. They built a hybrid architecture: high-recall OPF

detection + Qwen semantic re-judgment + wrapper policy decision. In the later stage, the team replaced traditional OCR tools with Qwen3-VL-8B-Instruct. It supports end-to-end multimodal processing of text, images and PDFs. This shift from laboratory models to engineering-ready services completed the project's core goals. It also maximized the practical value of the system.

8.2 Model Selection Basis and Underlying Tokenizer Principle

The project achieved excellent results mainly due to rigorous large model selection and reasonable design of the underlying tokenizer.

In model selection, the team abandoned two schemes. The first was pure rule-based matching. It cannot handle contextual ambiguity. For example, it may mistake a regular order number for a Tax File Number (TFN). The second was direct use of closed-source large models such as GPT-4. These models bring cross-border data compliance risks and extremely high inference latency.

The reason for choosing the hybrid architecture is clear. OPF is a lightweight sequence labeling model trained for privacy scenarios. It ensures zero missed detection of potential sensitive spans. Qwen3.5-9B has strong semantic understanding and can accurately disambiguate 80 types of PII. The team only trained a 1.3MB linear classification head and froze the Qwen backbone. This keeps the model's strong representation ability while limiting inference latency to 174–188 ms. It completely solves deployment challenges.

For data processing, the system uses a Byte-level BPE tokenizer. Different from traditional methods, it converts all text into UTF-8 byte streams. This design has two major advantages. First, it fully solves the out-of-vocabulary (OOV) problem. It is highly robust to Australian-specific identifiers and noisy OCR text. Second, it aligns the underlying sequences of OPF and Qwen. This reduces span boundary misalignment in cross-model processing and supports the 0.976 overlap recall rate.

8.3 Application Value and Academic Significance

Practical Application Value

This project directly solves pain points in Australian privacy governance and university document management. The final system adds a three-way risk decision mechanism: auto-redaction, manual review, and ignore. This design follows the

asymmetric risk principle of privacy protection. The cost of missing real PII is much higher than reviewing ambiguous samples.

The system is trained entirely on synthetic data and soft labels generated by the teacher model. It avoids legal and ethical risks of real personal data leakage and complies with current Australian privacy regulations. It uses a lightweight modular FastAPI backend and supports Docker deployment. It can be used as a pre-filtering component in university archiving, medical record de-identification, and government data sharing.

Academic and Technical Value

This project proves that in data privacy, freezing the large model backbone plus a tiny-parameter classification head can match the performance of full-parameter fine-tuning. This works when combined with proper hard-negative sample training. The project also uses the teacher model Qwen-27B to generate soft labels. It helps the student model learn asymmetric category distribution and confidence calibration. This provides a low-cost knowledge distillation method for domain entity extraction in resource-limited environments.

8.4 Filling Research Gaps

This project fills an important gap in privacy-preserving NLP. The industry currently lacks a lightweight, deployable redaction pipeline specially designed for Australian scenarios. We have found many flaws in mainstream privacy protection methods. Many teams choose common NER models. Nowadays, nearly all mainstream models have obvious drawbacks we cannot ignore. Their recognition ability drops sharply when dealing with finely divided local identity labels. Many teams turn to expensive closed-source large models, yet these tools do not perform well in real-world use. They frequently produce wrong results. Worse still, they tend to overinterpret ordinary content and label it as sensitive information.

We encountered quite a lot of practical troubles during project development. Our team gradually worked out a practical solution to fit real usage scenarios. We combined hard-negative sample learning with context-free inference evaluation. This approach stops the model from mechanically memorizing simple character combinations. After optimization, the system can easily tell real private data apart from regular numbers with similar appearances. Looking through past research in the

industry, few studies have adopted these techniques in such a complete and standardized way.

Data set problems troubled us a great deal in the early stage. The test data we used back then shared too many similarities. This pushed up the final scores artificially and failed to reflect the model's true performance. To solve this issue, we redesigned the entire testing framework. We adopted group-key hashed blind testing for all assessments. After multiple rounds of adjustments and improvements, the reliability of model evaluation across this field has been greatly lifted.

8.5 Practical Tips for Deployment

Based on our full experience from research to formal launch, we summed up a set of practical suggestions:

Simplify and Refine the Architecture

For privacy detection targeting specific scenarios, do not rely solely on end-to-end generative large models. A mixed structure works better here. We place a lightweight detector at the front end to secure a high recall rate, and connect it with a semantic classification head whose backbone is frozen. This matching method balances recognition accuracy, operating cost and response speed. The overall performance becomes far more stable.

Data Augmentation and False Positive Control

Items such as salary records and standard serial numbers often confuse the model and lead to misjudgments. When building training datasets, be sure to add sufficient hard-negative samples and regular data with no sensitive content. We train the model to understand meanings from context, instead of letting it just memorize fixed number formats.

Tokenizer Selection and Fault Tolerance Design

We apply Byte-level BPE tokenizer uniformly for formal deployment. It works compatibly with mixed-language texts, disordered raw data and noise generated from OCR results. Meanwhile, it can pinpoint the boundary of private content segments accurately.

Deployment Strategy and Human-Machine Collaboration

It is unnecessary to pursue full automation after the system goes online. We take risk isolation as the core principle and adopt a human-machine cooperation mode. Our team built an independent policy management module. We set strict confidence thresholds for all high-risk content. Once the system cannot reach a definite judgment, all uncertain results will be delivered to staff for manual review.

This working mode keeps data security well protected, and also maintains efficiency at a reasonable level.

9. LIMITATIONS AND FUTURE WORKS

We have now developed a usable pipeline, but it can still be further optimized in terms of accuracy, efficiency, generalization ability and real deployment. However, for future sustainable development, we can continue to improve this system in several directions.

Firstly, there is an improvement in model performance and data coverage. Currently, the system can identify many types of PII in Australian scenarios, such as TFN, Medicare number, student ID, mobile phone number, address, and bank information, etc. However, in some categories, such as gender, latitude and longitude, license plate number, some financial codes, the model may still have missed detections or false detections. In the future, more training samples can be added for these weak categories, especially by including more hard negative examples, to enable the model to better distinguish genuine PII from ordinary numbers.

The second direction, I think we can later develop a more detailed and specific set of security rules based on different contents. Currently, this system mainly relies on the credibility of the content, the context, and the risk score to decide whether to re-edit, conduct manual review, or simply ignore. Our model actually needs to identify 79 dimensions of information, some of which are closely related to the user and belong to highly sensitive content. For this type of information, we need to use more accurate and effective strategies so that the model can fully capture it upon identification. This way, less effort will be needed for manual review. As for those contents that have less relation to the user's security, the judgment criteria can be relaxed. The system first screens out the risky ones, and the low-risk ones or those that the machine is prone to

misjudge will be handed over to manual review. This way, both security and cost and efficiency can be better controlled, and both aspects can be taken into consideration.

The third direction is to enable the system to handle various types of files more effectively and enhance its multimodal recognition capabilities. Currently, we can handle images, scanned documents, and various PDFs. Generally, we first scan them using the VLM model, then extract the text, and subsequently extract the user's personal information from it. I believe this area can still be further optimized and upgraded. Since the file formats uploaded by users are not standardized, some are very complex tables, some have very messy layouts in PDFs, there are handwritten content images, and even files with very low resolution and blurry images. We plan to further enhance the model's own capabilities so that when it encounters these irregularly formatted file inputs, it can also stably and accurately extract all the user's relevant information, ensuring that the text is not lost or missed, and the recognition results are more reliable.

The fourth point is to optimize the performance and deployment. This system was originally designed for the internal staff of the school. Although the current hybrid model we use has saved a lot of computing power compared to directly using a large model and has also made the response faster, I still think there is room for further optimization. Later, more adjustments can be made within the system to enable it to handle more requests simultaneously, run more smoothly, and respond faster. Additionally, the usage permissions of different users are different. We will also improve the API authentication system and formulate more detailed security strategies to ensure that the system is stable, reliable, and secure after its official launch. At the same time, a complete log recording function will be implemented, allowing the tracking of who operated what, ensuring full traceability throughout the process, and further safeguarding security.

In summary, the future upgrade directions we have considered for this project system in our company mainly include the following aspects: First, continue to optimize the data construction to enable the model to learn more accurate confidence information; Second, ensure local deployment and performance optimization, responding to user needs faster while maintaining security; Third, enhance the model's capabilities so that it can better understand various complex and non-standard images,

PDF files, etc., and extract more complete information; Fourth, continuously update security strategies, formulating different handling rules at different levels based on the security level of the information.

Overall, we focus on data, model capabilities, security strategies, and system performance, making the entire system more stable, accurate, and user-friendly.

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